

FREQUENCY BRANCH · FREQCNN + FFT LOG-MAGNITUDE · SIDANG STUDY GUIDE

Near-random, by **measurement.**

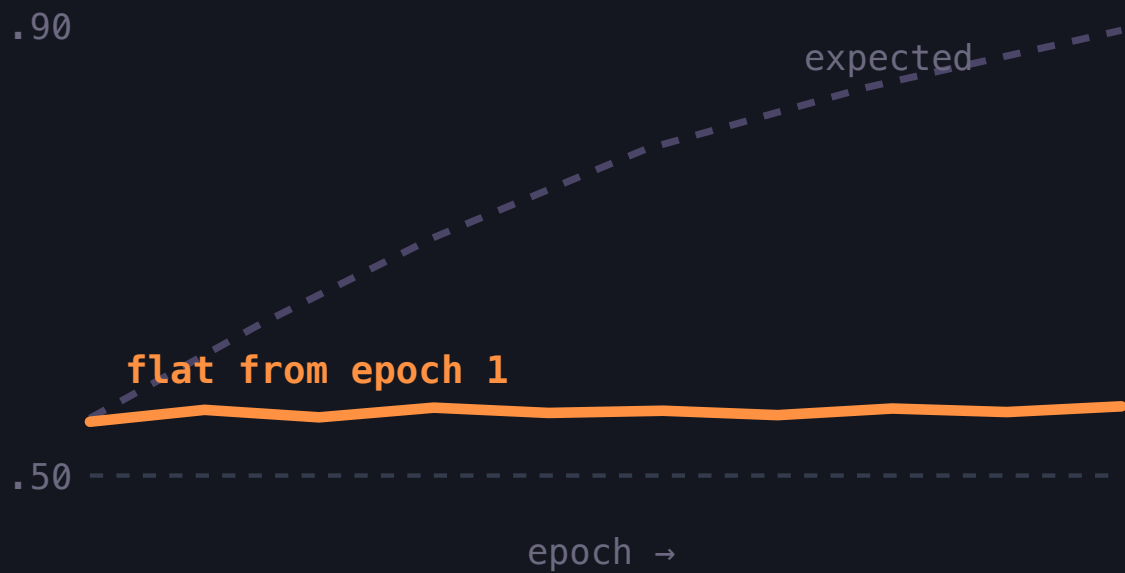
You own the frequency branch: the FFT log-magnitude map, the **FreqCNN**, and the thesis's **central negative result**. Frequency is the variable this whole comparative study measures, and it being the weakest detector is the **result**, not a bug. Two of the hardest reviewer targets land on you.

§0 · THE ONE-LINER YOU DEFEND

The frequency branch is our **weakest detector** — in-dataset AUC only **~0.56** on both datasets, and its validation AUC is **flat from epoch 1**. That is not a training failure. We ruled out under-powering three ways (more data, a bigger backbone, verified preprocessing) and near-chance survives all three. So the bottleneck is the **information content of the compressed, face-cropped FFT input**, not the model. A **controlled falsification** of the assumption that spectral artifacts are a strong, general cue.

0.562

IN-DATASET AUC · FFPP & CDF · N=750
THE WEAKEST MODEL ON BOTH DATASETS



validation AUC – it never found signal (Gambar 4.10/4.11)

§1 What to study — the ordered curriculum

Each item: *what* · *why you need it* · *what to read*. Tick them off. Priority and time to budget are marked on every card.

1

The FFT transform, exactly

image_to_fft_logmag · fft_utils.py:22-33

~1.5 HR

CORE

This is the “explain your frequency representation” question. You must walk it step by step.

- **Grayscale → resize 224×224** (`img.convert("L")`). Color is dropped; the branch sees luminance structure only.
- `np.fft.fft2` → `np.fft.fftshift` — 2-D Fourier transform, then shift the **DC (zero-frequency)** term to the image center. After shift: **center = low freq**, **edges = high freq**.
- **Magnitude only** (`np.abs`) — keeps spectral **energy** per frequency. **Phase is thrown away** (memorize this — failure cause #2).
- **Gaussian high-pass mask** (`_highpass_mask` , cutoff 0.15) — attenuates the dominant low-frequency center so the CNN focuses on the high-frequency band where GAN up-sampling fingerprints live.
- `log1p` = `log(1+|F|)` — log scaling compresses the huge dynamic range of the spectrum into a trainable map.
- Output: a **single-channel 224×224** FFT log-magnitude map, cached as `.npy`.

ONE-SENTENCE ANSWER

“We take each frame to grayscale, compute its 2-D FFT, keep the magnitude spectrum, apply a Gaussian high-pass to emphasize high frequencies, and log-scale it — producing a single-channel FFT log-magnitude map. The FreqCNN learns discriminative features from that map.”

2

The high-pass mask is *soft*, not a hard cut

`_highpass_mask` · `fft_utils.py:9-19`

30 MIN

CORE

A likely trap: “you said low frequency is unused.” **Wrong — it's attenuated, not removed.**

- Formula: $H = 1 - \exp(-d^2/2\sigma^2)$, with $\sigma = \text{cutoff} \cdot \text{size} = 0.15 \cdot 224 \approx 33.6 \text{ px}$. d = distance from center.
- **Only the pure DC term ($d=0$) is multiplied by ~ 0 .** At $d=\sigma$ the mask ≈ 0.39 , at $d=2\sigma \approx 0.86$, edges $\rightarrow 1.0$.
- **Why soft, not hard:** (a) a smooth roll-off avoids **ringing** artifacts a sharp cut would introduce, and (b) some discriminative signal (spectral roll-off, Durall [8]) lives in the **mid** band, not only the edge.
- **Correct phrasing:** “high frequencies are **amplified**, low frequencies are **suppressed (attenuated)** — not discarded.” (The thesis says “*menekan dominasi*” — suppress, not remove.)

3

Why FFT *should* expose deepfakes — the literature you are testing

45 MIN

CORE

- GANs up-sample with **transposed convolution / interpolation**, which inserts zeros $\rightarrow$ replicates the spectrum into the high-frequency band $\rightarrow$ periodic **checkerboard** grid artifacts (Odena [21]).
- CNN-based generators **fail to reproduce the natural spectral distribution** — their power spectrum deviates from real images, mostly in the **high-frequency tail** (Durall [8]; Zhang [9]).
- Frequency-aware detectors exploit this: F^3 -Net (Qian [11]), DCT anomalies (Giudice [10]), FreqNet (Tan [12]).
- **Your study asks:** does a *simple raw-FFT branch* capture this and help fusion? **Answer, under our conditions: no.** You're testing the claim, not assuming it.

- **FreqBlock** (the building block): `Conv3x3 → BatchNorm → ReLU`, plus a **residual shortcut** (a `1x1` conv when channels change, else identity), added **before** a `MaxPool2d(2)`. So one block = `pool(conv(x) + shortcut(x))`. Residual idea from ResNet (He [5]).
- **Configurable depth/channels** via `freq_depth / freq_base_channels`. ▲
THE RESULTS RUN USES `DEPTH=5, BASE_CHANNELS=64` (`config.yaml:22-23`), **not** the constructor defaults (3/32).
 - Channel progression `1 → 64 → 128 → 256 → 512 → 512`, feature dim **512**, ~**4.2M params** (4,217,217).
 - The docstring's "depth=5 → 256-d, ~700K" describes base=32; that is **not** what the experiments use. Say **512-dim, 4.2M**.
- `.features` = conv blocks + `Dropout2d(0.2)` + `AdaptiveAvgPool2d(1)` → a 512-vector. `.classifier` = `Flatten → Linear(512→256) → ReLU → Dropout(0.3) → Linear(256→1)`.
- **Key fact for the hybrid:** inside the hybrid, only `FreqCNN.features` is used — the classifier head is **bypassed** (`hybrid_fusion.py:58`). Standalone `freq` uses the full net.
- Input **224×224, single channel**. Layer-by-layer is in **Tabel 3.7**, diagrams in **Gambar 3.8/3.9**.

5

FFT normalization + augmentation — the preprocessing you must defend

`deepfake_data.py:129–141`

45 MIN

- **Per-dataset z-score:** $(x - \text{mean}) / \text{std}$ from `fft_stats.json`. Auto-computed after `compute_fft_cache.py`. FFPP stats: **mean ≈ 5.84 / std ≈ 1.28** (the live file; docs sometimes quote 5.78 — code reads the file, so 5.84 runs). The old fallback `std=3.0` was **$\sim 2.3\times$ too big** — a real bug that was found and fixed.
- **Train-only FFT augmentation:** Gaussian noise `$\sigma=0.05$` added to the map, plus **5%-probability spectral-band masking** (zero a random horizontal/vertical frequency band) to stop the net memorizing any single band. Off at inference.

In-dataset AUC (mean over seeds 0/1/2)

	N=250	N=500	N=750
FFPP freq	0.469	0.545	0.562
CDF freq	0.500	0.549	0.562
FFPP spatial (contrast)	0.743	0.693	0.778
CDF spatial (contrast)	0.914	0.945	0.971

Freq is **last on both datasets at every reliable tier**. 3× the data (250–750) shifts AUC ~0.06 then plateaus at ~0.56. Validation-AUC curve is flat from epoch 1 (BAB IV §4.1.7, Gambar 4.10/4.11) – it never found signal.

Cross-dataset, n=750 (AUC / recall)

DIRECTION	AUC	RECALL
FFPP→CDF freq	0.606	0.064 freq's own collapse
CDF→FFPP freq	0.575	0.531

Mirror of spatial: freq **collapses FFPP→CDF** (recall 0.064) while spatial collapses CDF→FFPP. Same mechanism (score distribution sinks past the fixed 0.5 cutoff) – a **threshold-calibration artifact under domain shift**, not a total loss of ranking. See `RECALL_COLLAPSE_ANALYSIS.md`.

Your core defense (Slide 19, BAB IV §4.2.4). Memorize all four, **in order of strength**.

01 Preprocessing destroys the artifacts (strongest). MTCNN face-crop removes edge/background context carrying spectral traces, **and** c23 compression is a low-pass filter that kills exactly the high-frequency band where GAN fingerprints live (Mejri [31]). Durall got ~100% on **raw, uncompressed** output; we use cropped, c23 frames. **Visible proof:** real vs fake FFT spectra look indistinguishable (Gambar 4.2).

02 Phase is discarded. Magnitude only. Phase carries most structural info (Oppenheim & Lim [25]); SPSL (Liu [26]) shows phase catches up-sampling artifacts invisible in magnitude. A **deliberate, flagged** handicap → future work.

03 CNN spectral / texture bias. CNNs learn texture and **low** frequencies first (Geirhos [28]; Rahaman [29] "F-principle"; Wang [30]). A weak high-frequency signal is intrinsically hard for a shallow CNN without architectural help.

04 Over-simplified representation. A single **raw FFT magnitude map** into a **shallow CNN** is less expressive than SOTA's learned/decomposed frequency spaces — DCT (Giudice [10], Qian [11]), FreqNet (Tan [12]). → future work.

● THE FRAMING THAT WINS

"This is **not a contradiction of the literature, it's a boundary condition.** Spectral artifacts are real; *exploiting* them is **conditional** on representation, fusion design, and test difficulty. We mark exactly *when* frequency stops helping."

8

Proving it's NOT under-powered — the rigor that upgrades “failure” to “finding”

45 MIN

CORE

A reviewer's sharpest attack is “your freq net is just too small / under-trained / mis-normalized.” You dismantle each.

■ THE NOT-UNDER-POWERED TRIO

- **Sample-size sweep = a direct power analysis.** If freq were data-starved, AUC would keep rising with n . It **plateaus at ~ 0.56** . ($n=100$ is noisy — argue the $250 \rightarrow 750$ trend.)
- **Architecture control (FreqCNN vs ResNet18).** A standard **ResNet18** (11.2M params, **$2.6\times$ bigger**), both scratch and pretrained — every arm still sits at **AUC 0.51–0.63** (`FREQ_BENCHMARK_RESULTS.md`). Bigger backbone can't lift FFT input out of near-chance $\rightarrow$ bottleneck = **input information**, not capacity.
- **Verified preprocessing.** FFT stats recomputed (fixed the $\text{std}=3.0$ bug); splits **by video ID**; identical optimizer/schedule/epochs/early-stopping as spatial. Apples-to-apples, 3 seeds, small std ($\sim 0.007\text{--}0.015$).

■ HONEST CAVEAT

Cite the **scratch** arm as the clean win (FreqCNN wins 4/4 under matched training). The **pretrained** ResNet18 arm is a **2–2 split** — do **not** claim it as a FreqCNN victory.

Ranked by likelihood. Answer honestly; frame negatives as findings. **Try to recall each answer before revealing it.**

REVEAL ALL

FQ1 "Explain your frequency representation — walk me through the FFT step."

+

ALMOST CERTAIN

FQ2 "Your frequency branch is near-random (AUC 0.56). Isn't that just a failed model?"

+

CERTAIN · THE HARD ONE

FQ3 "Why does frequency work near-perfectly in papers like Durrall but not for you?"

+

CERTAIN

FQ4 "Why magnitude only? You cite Oppenheim that phase carries more information."

+

LIKELY · THE PHASE TRAP

FQ5 "Walk me through FreqCNN. Why this design and not just a big CNN?"

+

LIKELY

FQ6 "Frequency looks inverted at $n=250$ (AUC $0.469 < 0.5$). Is your labeling broken?"

+

A TRAP

FQ7 "Recall for freq FFPP→CDF is 0.064 — total collapse?"

+

LIKELY

FQ8

"What does the high-pass filter actually remove?"



POSSIBLE

FQ9

"If frequency is useless, why include it at all / why isn't this a wasted branch?"



POSSIBLE

FQ10

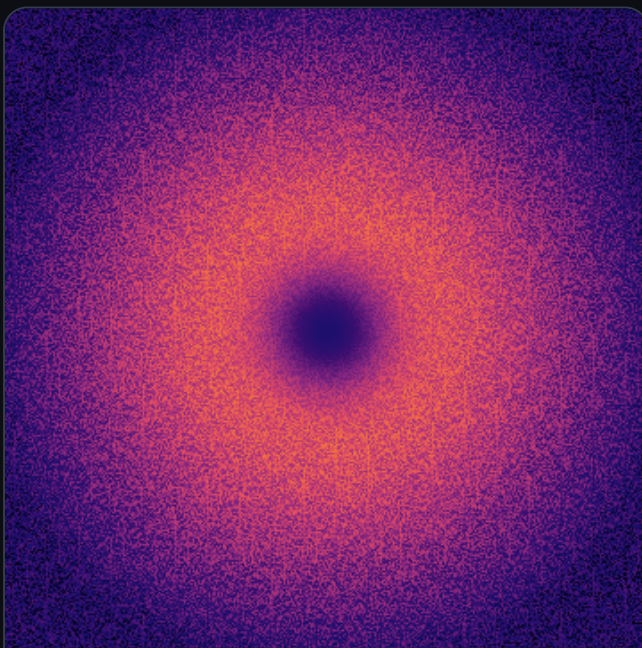
"Why single-channel / grayscale? Don't you lose the color artifacts?"



POSSIBLE

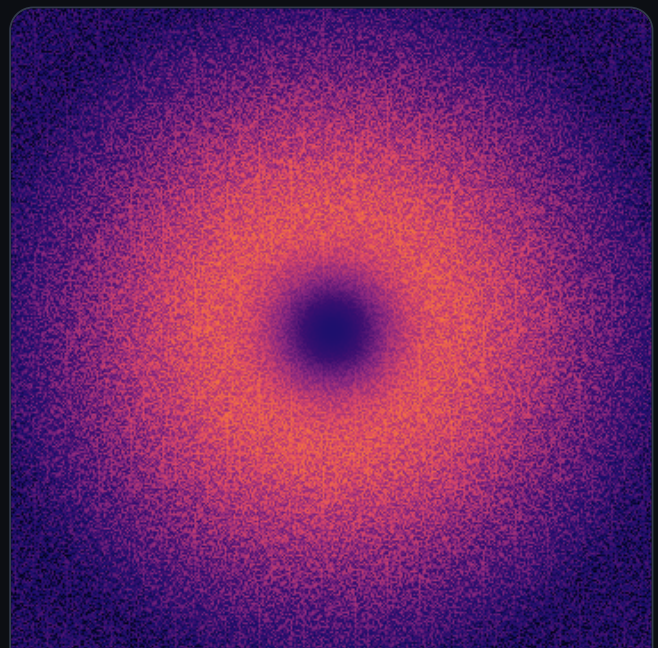
§2b The visible proof — real vs fake spectra

Your headline visual (Gambar 4.2 / the demo's MAGMA spectra gallery). Point at it and say it out loud.



REAL

FFT log-mag · magma



FAKE

FFT log-mag · magma

● THE ONE SENTENCE FOR THIS PANEL

"Real and fake spectra are essentially **indistinguishable** under crop + c23 — that's the human-visible reason the frequency branch carries little signal, and exactly what its ~0.56 AUC reports."

§3 Frequency-branch numbers card

Keep this on your phone. Every value quoted from the results run.

FreqCNN · FFT log-magnitude

RESULTS RUN

REPRESENTATION	grayscale → FFT2 → fftshift → magnitude → Gaussian high-pass (cutoff 0.15) → log1p → 1×224×224 map
PHASE	Discarded (magnitude only) — deliberate, flagged limitation
MODEL	FreqCNN — residual <code>FreqBlock</code> (Conv3×3→BN→ReLU + 1×1 shortcut → MaxPool)
CONFIG (RUN)	freq_depth=5, freq_base_channels=64 → channels 1→64→128→256→512→512, feature dim 512
PARAMS	~4.2M (4,217,217) — vs Xception 20.8M
NORMALIZATION	per-dataset z-score, FFPP mean ≈ 5.84 / std ≈ 1.28 (file); old fallback std 3.0 was ~2.3× too big
AUG (TRAIN)	FFT Gaussian noise $\sigma=0.05$ · 5%-prob spectral-band masking
IN-DATASET AUC	n=750 — FFPP 0.562 · CDF 0.562 (weakest model, both)

CROSS-DATASET	n=750 — FFPP→CDF AUC 0.606 (recall 0.064) · CDF→FFPP AUC 0.575 (recall 0.531)
THRESHOLD (DEMO)	0.451 (Youden's J on FFPP val — near 0.5, consistent with near-random)
POWER CHECKS	data sweep plateaus ~0.56 · ResNet18 (2.6× params) still 0.51–0.63 · 3 seeds, small std
KEY CITATIONS	[8] Durall · [9] Zhang · [21] Odena · [11] Qian/F ³ -Net · [10] Giudice · [12] Tan/FreqNet · [25] Oppenheim&Lim · [26] SPSL/Liu · [28] Geirhos · [29] Rahaman · [30] Wang · [31] Meiri · [5] He/ResNet

§4 If you only have one evening

1 Walk the FFT transform end to end (FQ1) and know the high-pass is **soft** (FQ8).

2 Memorize freq AUC ~0.56 + the flat validation curve and the **four failure causes with citations** (FQ2, FQ3).

3 Rehearse the “not under-powered” trio — data sweep, ResNet18 control, verified normalization (FQ2, §1.8).

4 Own the phase handicap as deliberate future work, not an oversight (FQ4).

5 Re-read `sidang_study_notes_detailed_2026-06-25.md` Steps 03 & 07

and `FREQ_BENCHMARK_RESULTS.md` once.

§5 Literature notes worth keeping

■ BEFORE SIDANG

In-text numbers follow the docx. **Verify each [N] against Daftar Pustaka** — the fact-check flagged in-text / bibliography desync.

Durall et al. [8] — “Watch your Up-Convolution”

GAN up-sampling fails to reproduce the spectral distribution; deviation is in the **high-freq tail** → ~100% on **raw** output. Your **justification** for trying FFT *and* the **contrast** that explains your weak result (they used uncompressed; you use c23 crops).

Odena [21] / Zhang [9]

Checkerboard & spectral artifacts from transposed conv — the mechanism your branch was designed to catch.

Mejri [31] — “Leveraging High-Frequency Components”

Compression/filtering destroys high-frequency deepfake cues. **The single best citation for failure cause #1.**

Oppenheim & Lim [25] + Liu/SPSL [26]

Phase carries structure; phase surfaces up-sampling artifacts — back the phase-as-future-work argument. Oppenheim is about **perceptual structure**, not artifact detection — don't let it become “magnitude is worthless.”

Geirhos [28] · Rahaman [29] · Wang [30]

Texture bias / F-principle / high-freq ⇔ generalization — the CNN-bias reason a shallow FreqCNN struggles on a weak high-freq signal (failure cause #3). The “teori tambahan BAB II” added 2026-06-17.

Qian/F³-Net [11] · Giudice/DCT [10] · Tan/FreqNet [12]

Richer frequency-aware SOTA — frame your raw-FFT-magnitude as the **simple baseline** and these as the future-work direction.

He/ResNet [5]

The residual idea inside `FreqBlock`.

SpecXNet (Alam [13]) · FSBI (Hasanaath [16])

Dual-domain / freq-enhanced related work — cite as **inspiration**, and be precise that they do **not** append FFT as a 4th channel (that's early fusion, which you did not evaluate).

§6 Known quirks to OWN before they're pointed out

■ DOCSTRING VS RESULTS CONFIG

`freq_cnn.py`'s docstring says depth=5 → 256-d ~700K; that's for **base=32**. Your run is **base=64 → 512-d, 4.2M**. Quote the config values, not the docstring.

■ .FEATURES ONLY IN HYBRID

The FreqCNN's own classifier head is **bypassed** inside the hybrid; only the 512-d feature vector is used. Standalone `freq` uses the full head.

■ FFT-STATS NUMBER

File on disk is **5.84 / 1.28**; some docs say 5.78. Code reads the file live → 5.84 runs. Quote 5.84.

■ TWO FFT RE-ENCODES DIFFER BY ONE JPEG HOP

Training computes FFT from the saved JPEG crop, the demo from the in-memory crop — immaterial, but be honest if asked.

■ FREQ THRESHOLD $0.451 \approx 0.5$

That near-0.5 operating point is itself *evidence* of near-randomness (no confident cutoff exists), not a tuning mistake.

§7

The demo — freq-branch things a reviewer can point at live

Pair with `sidang_study_guide_DEMO_debug.md` (the full operational guide). This is **only** the frequency-specific angles. The demo runs all three models side-by-side so the negative result is *visible* — your branch is the one that's supposed to look weak, and it does.

What is *yours* on screen

- The “**FFT spectra**” gallery (`inference._build_visuals` → `_spectrum_to_rgb`, MAGMA colormap) is **exactly the freq/hybrid input** (raw `image_to_fft_logmag`, pre z-score). Your headline visual proof (Gambar 4.2).
- The **freq verdict card** shows the fake-probability bar with a tick at the **0.451** threshold.

Freq-specific failure modes

SYMPTOM	CAUSE	WHAT TO SAY
Freq card near its threshold / flips on tiny changes	freq is near-random (AUC ~ 0.56), threshold $0.451 \approx 0.5$	"Expected — a near-random model has no confident operating point. Evidence, not a defect."
Cards show "⚠ FFT calibration estimated"	<code>fft_stats.json</code> missing → fallback mean 5.0/std 3.0 (<code>inference.py:123-133</code>)	Copy <code>outputs/fft_cache/FFPP/fft_stats.json</code> → <code>checkpoints/</code> . Spatial is unaffected (RGB-only).
Freq says the opposite of spatial on an OOD clip	freq $\approx$ coin flip; clip is off-distribution	"Freq alone is near chance, so disagreement is uninformative — that's the finding, and why we don't ship freq alone."
Freq is fast but weak	1-channel FFT input, tiny 4.2M net	latency isn't the point; the point is the flat spectra above it

Reviewer: "change X and show me" — freq edit points

- "Turn off the high-pass" → `image_to_fft_logmag(..., highpass=False)` in `inference._preprocess`. **▲ SAY FIRST: THIS DIVERGES FROM TRAINING, SO THE NUMBERS STOP MATCHING THE THESIS.**
- "Change the FFT high-pass cutoff" → `_highpass_mask(cutoff=...)` in `src/fft_utils.py`. Same caveat.
- "Run just the frequency model" → in `inference.py` `MODELS_SPEC`, keep only the `("freq", ...)` row.

● THE ONE SENTENCE FOR THE DEMO

"The FFT-spectra panel *is* my defense: you can see for yourself that fake spectra look like real ones after cropping and compression — so the frequency branch has almost nothing to grab, and that's exactly what its ~ 0.56 AUC reports."

COMPANION GUIDES

◆ FREQUENCY branch

SPATIAL branch

HYBRID branch

DEMO debug

Sidang study guide · FreqCNN + FFT · deepfake_hybrid · values from the results run